

### 1.what is UAT ? who will do the testing and what is the purpose of testing?

**UAT (User Acceptance Testing)** is the **final phase of the testing process** where the **end users or clients** validate whether the software meets their **business requirements** and is ready for real-world use.

### 2.what is the difference between functional and non-functional testing?

| Functional Testing                                        | Non-functional Testing                                          |
|-----------------------------------------------------------|-----------------------------------------------------------------|
| It verifies the operations and actions of an application. | It verifies the behavior of an application.                     |
| It is based on requirements of customer.                  | It is based on expectations of customer.                        |
| It helps to enhance the behavior of the application.      | It helps to improve the performance of the application.         |
| Functional testing is easy to execute manually.           | It is hard to execute non-functional testing manually.          |
| It tests what the product does.                           | It describes how the product does.                              |
| Functional testing is based on the business requirement.  | Non-functional testing is based on the performance requirement. |

### 3.load testing will come under functional testing or non-functional testing?

And why it is coming under that category?

Load testing checks how the application performs when many users access it at the same time. Since it measures performance and not functionality, it is categorized under non-functional testing.

### 4.what is exploratory testing and when we will do this testing?

**Exploratory Testing** is a **simultaneous process of learning, test design, and test execution**, done when requirements are unclear, time is limited, or when testers want to discover hidden defects through creative, experience-based testing.

5. what is black box testing and how it differs from white box testing?

> **Black Box Testing = Test functionality without seeing code**

> **White Box Testing = Test internal logic with full code visibility**

> Both are essential for complete software quality.